Canadian Driver's Licence OCR Reference

Oct 11, 2026 · @Jjv

Overview

Build the extractor around Ontario as the baseline, then treat every other jurisdiction as a set of differences: licence-number pattern, date format, label wording and language, field order, and what the back of the card encodes. This doc covers all 13 issuing jurisdictions (10 provinces, 3 territories) for standard driver's licences.

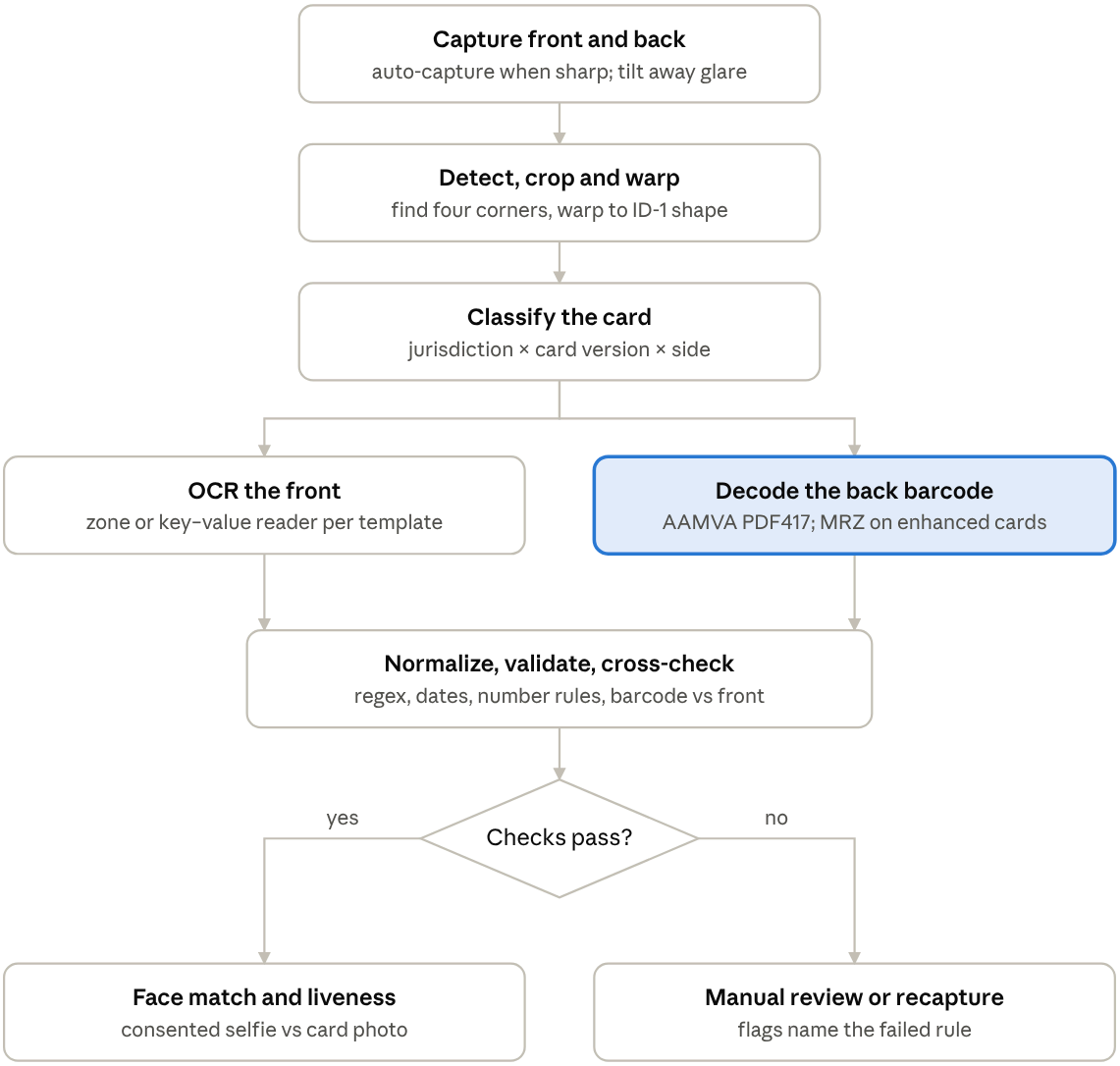
Every card picture here is a schematic field map drawn for developers, not a copy of a real licence. Zone positions are indicative only; calibrate them against each jurisdiction's official specimen (linked in Sources) and real, consented captures before training. All sample values are fictitious.

How to read the schematics:

- **Blue** = name
- **Green** = dates (birth, issue, expiry)
- **Purple** = licence data (number, class, conditions, endorsements, reference)
- **Orange** = address
- **Cyan** = physical descriptors (sex, height, weight, eyes, hair)
- **Grey** = photo, secondary photo, signature
- **Slate** = issuer title and machine-readable zones (barcodes, magnetic stripe)
- **Pink** = data to mask and never store (health number, citizenship marker)
- **Light grey** = printed text OCR can skip
- **Red dashed outline** = differs from Ontario
- **Grey dotted outline** = no public source confirms it; verify on a specimen

Extraction pipeline

Decode the barcode first; OCR fills in and cross-checks the front



verification pipeline · 8 steps, 1 decision

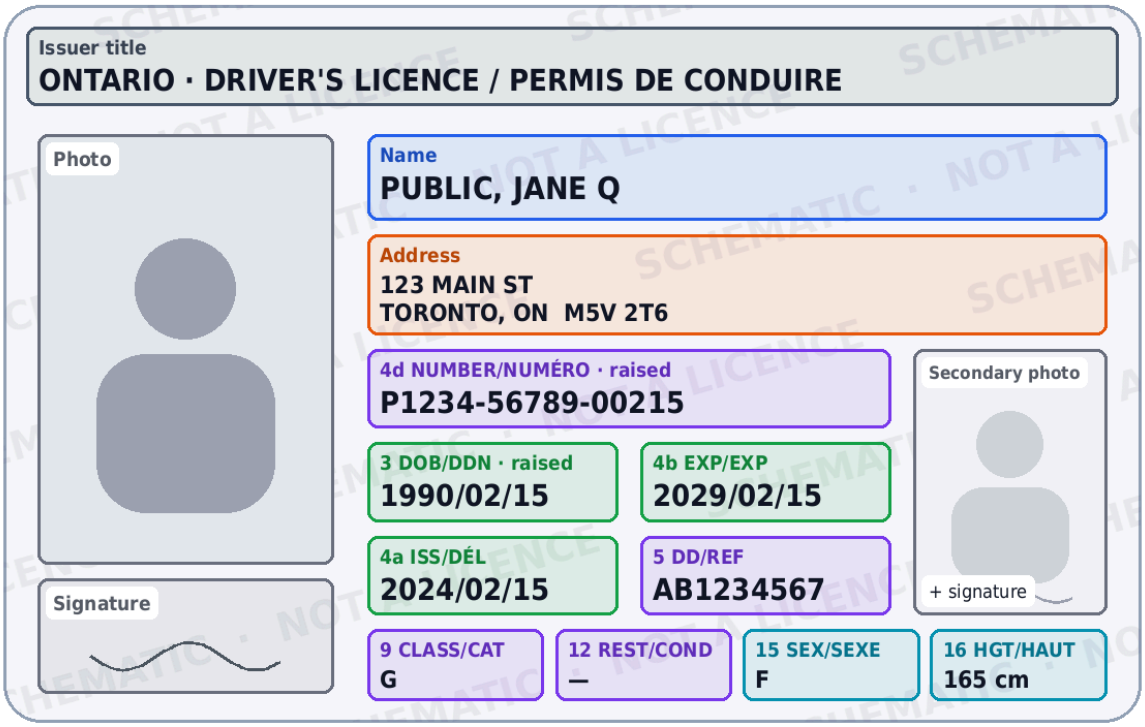
Run OCR and barcode decoding side by side once the card is classified. A disagreement between them, or a failed number rule, sends the submission to review with the rule named instead of rejecting it outright.

Ontario baseline

Ontario's current card is bilingual (English/French), prints every date as YYYY/MM/DD, and carries a machine-readable barcode on the back with the licensee's data (AGCO, field table). Vendors list it as the 2011 design, plus a 2010 enhanced licence (Incode).

Ontario · front (baseline)

Schematic field map · not a licence · current design (vendor list: 2011) · positions indicative · fictitious values



OCR notes

- Labels are bilingual short forms; confirm exact wording on the AGCO specimen before using them as anchors.
- Licence number and birth date are raised: expect shadows under angled light.
- Cards issued at ages 16–18 add an AGE 19 banner with a date near the bottom, beside the birth date.
- No eye colour, hair colour or weight on this card.

Issuer title (classifier cue) · Name · Dates · Licence data · Address · Physical descriptors · Photo / signature

Ontario · back (baseline)

Schematic field map · not a licence · current design · positions indicative · fictitious values



OCR notes

- Decode the barcode first: AGCO notes it carries licensee data, so it is the reference for cross-checking the front.
- Map DAQ, DCS, DAC, DBB, DBD, DBA and DCF into the normalized record.
- Ask for a closer capture of the back than the front: PDF417 needs sharp, glare-free modules.

Machine-readable Not needed for OCR

Field	Expected printed label	Fictitious example	Format / validation	Notes
Name	none (name block)	PUBLIC, JANE Q	Uppercase; accents ç è é ê ë allowed	Surname first expected; confirm the separator on the specimen
Address	none (address block)	123 MAIN ST, TORONTO, ON M5V 2T6	Postal code `[A-Z]\d[A-Z] ?\d[A-Z]\d`	2–3 lines; province always ON

Field	Expected printed label	Fictitious example	Format / validation	Notes
Licence number	4d NUMBER/NUMÉRO	P1234-56789-00215	`^[A-Z]\d{4}-?\d{5}-?\d{5}$`	Raised text; rules in the next section
Date of birth	3 DOB/DDN	1990/02/15	YYYY/MM/DD	Raised text; cross-check with licence number
Issue date	4a ISS/DÉL	2024/02/15	YYYY/MM/DD	
Expiry date	4b EXP/EXP	2029/02/15	YYYY/MM/DD	Reject if in the past
Document reference	5 DD/REF	AB1234567	`^[A-Z]{2}\d{7}$`	New value on every reissued card; use it to spot re-submissions
Class	9 CLASS/CAT	G	One or more of A, B, C, D, E, F, G, G1, G2, M, M1, M2	
Conditions	12 REST/COND	(blank)	Short letter codes	Usually blank
Sex	15 SEX/SEXE	F	M, F or X	X is a valid designation
Height	16 HGT/HAUT	165 cm	`^\d{3} ?cm$`	Metric only
Photo	none	face image	—	Crop for face match
Signature	none	—	—	Not read by OCR
Secondary photo + signature	none	—	—	Bottom-right security feature

Field	Expected printed label	Fictitious example	Format / validation	Notes
AGE 19 banner	AGE 19 + date	AGE 19 2027/05/10	Date	Only on cards issued at ages 16–18
Barcode (back)	none	—	AAMVA PDF417	Primary machine-readable source

Ontario prints no eye colour, hair colour or weight; return null for those. The label wording is the expected bilingual form; confirm it on the AGCO specimen before you hard-code text anchors.

Ontario licence number anatomy

The Ontario number is 15 characters: one letter and 14 digits, printed as A1234-56789-01234 (Ontario design system). Three rules are published, so they are safe to validate (AGCO):

Characters (example P1234-56789-00215)	Value	Meaning	Validate?
Letter	P	First letter of the surname	Yes
Digits 1–8	1234 5678	Not published	No
Digits 9–10	9 and 0 (they straddle the second dash)	Last two digits of birth year (1990)	Yes
Digits 11–12	02	Birth month, per an insurer's published format (Sonnet); AGCO does not list it	Soft flag only
Digits 13–14	15	Day of birth (01–31)	Yes

Treat any failed rule as a review flag, not an automatic rejection: name changes and OCR misreads both cause mismatches. Some numbers start with the given name's initial because the names were recorded in reverse order, so check both initials before flagging (Sonnet).

```
import re, unicodedata

ON_RE = re.compile(r"^([A-Z])(\d{4})-?(\d{5})-?(\d{5})$")

def initial(name: str) -> str:
    return unicodedata.normalize("NFKD", name.strip().upper())[:1]

def check_on_number(number, surname, given_names, dob) -> list[str]:
    """Return failed checks; an empty list means every rule passes."""
    m = ON_RE.match(number.upper().replace(" ", ""))
    if not m:
        return ["format"]
    digits = m.group(2) + m.group(3) + m.group(4)  # 14 digits
    failed = []
    if m.group(1) != initial(surname):
        reversed_names = m.group(1) == initial(given_names)
        failed.append("names_reversed" if reversed_names else
"surname_initial")
    if digits[8:10] != f"{dob.year % 100:02d}":
        failed.append("birth_year")
    if digits[10:12] != f"{dob.month:02d}":
        failed.append("birth_month_soft")
    if digits[12:14] != f"{dob.day:02d}":
        failed.append("birth_day")
    return failed
```

Normalized output schema

Map every jurisdiction into one record so downstream code never branches on province. Dates are ISO 8601, the licence number is stored without separators, and each field records where it came from and how confident the reader was.

```
{
  "jurisdiction": "ON",
  "card_version": "ON-2011",
  "sides_captured": ["front", "back"],
```

```
  "licence_number": "P12345678900215",
  "licence_number_printed": "P1234-56789-00215",
  "family_name": "PUBLIC",
  "given_names": "JANE Q",
  "date_of_birth": "1990-02-15",
  "issue_date": "2024-02-15",
  "expiry_date": "2029-02-15",
  "sex": "F",
  "height_cm": 165,
  "weight_kg": null,
  "eye_colour": null,
  "address": {
    "lines": ["123 MAIN ST"],
    "city": "TORONTO",
    "province": "ON",
    "postal_code": "M5V2T6"
  },
  "licence_classes": ["G"],
  "conditions": [],
  "endorsements": [],
  "document_reference": "AB1234567",
  "field_sources": {"licence_number": "barcode", "date_of_birth":
"barcode+ocr"},
  "field_confidence": {"licence_number": 0.99, "family_name": 0.97},
  "checks": {
    "barcode_matches_front": true,
    "number_rules_pass": true,
    "not_expired": true
  }
}
```

- `sex` is M, F or X exactly as printed; never infer it.
- A field the card does not print is `null`, never an empty string.
- Never store a health number (PHN) or a citizenship marker, which some newer cards print; mask them at capture (see Privacy).
- Keep raw OCR text and barcode payloads only as long as your review queue needs them.

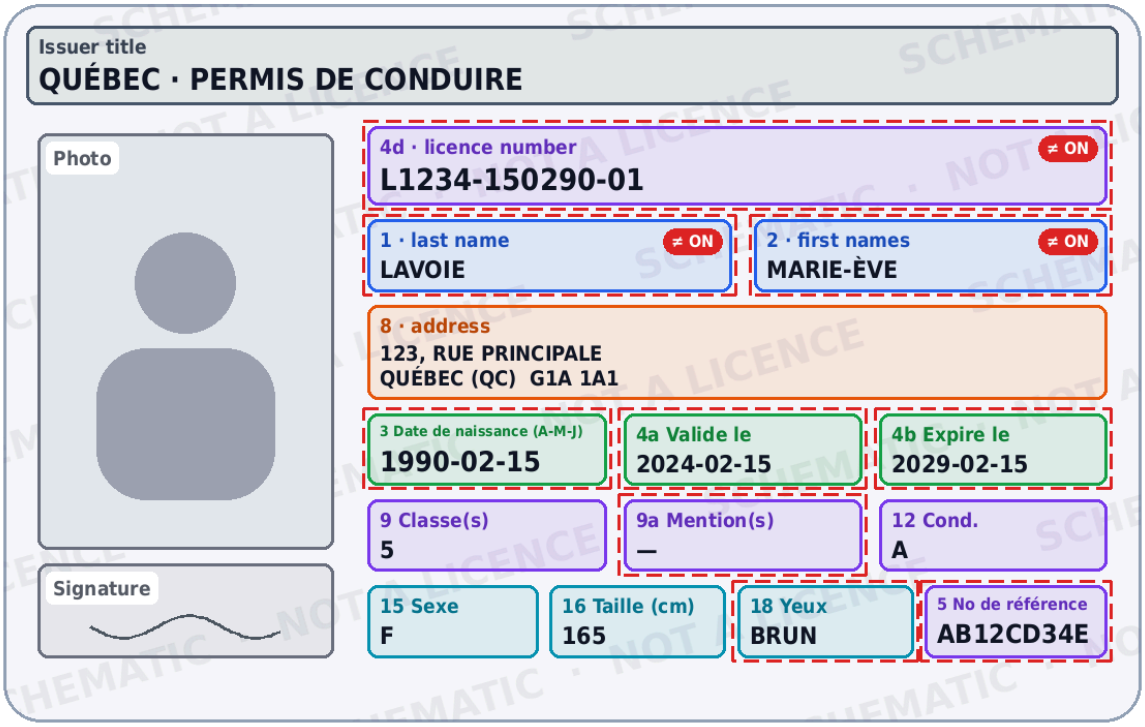
Province and territory differences

Each card is drawn against the Ontario baseline: red dashed outlines mark confirmed differences, grey dotted outlines mark what no public source confirms. Only Quebec, British Columbia and Alberta publish enough to place fields; the other cards are generic field inventories.

Quebec

Quebec · front

Schematic field map · not a licence · 2023 design, fields per SAAQ · positions indicative · fictitious values



What differs from Ontario

- French-only labels with numbered fields (1, 2, 3, 4a, 4b, 4d, 5, 8, 9, 9a, 12, 15, 16, 18): anchor on the numbers.
- Number L1234-150290-01 = surname initial + 4 digits + birth date as DDMMYY + 2 digits.
- Dates are YYYY-MM-DD (the label says A-M-J); surname and first names sit on separate numbered lines.
- Eye colour, endorsements (9a) and an alphanumeric reference number (5) are printed; Ontario prints none of them.

Issuer title (classifier cue) · Name · Dates · Licence data · Address · Physical descriptors · Photo / signature · Differs from Ontario

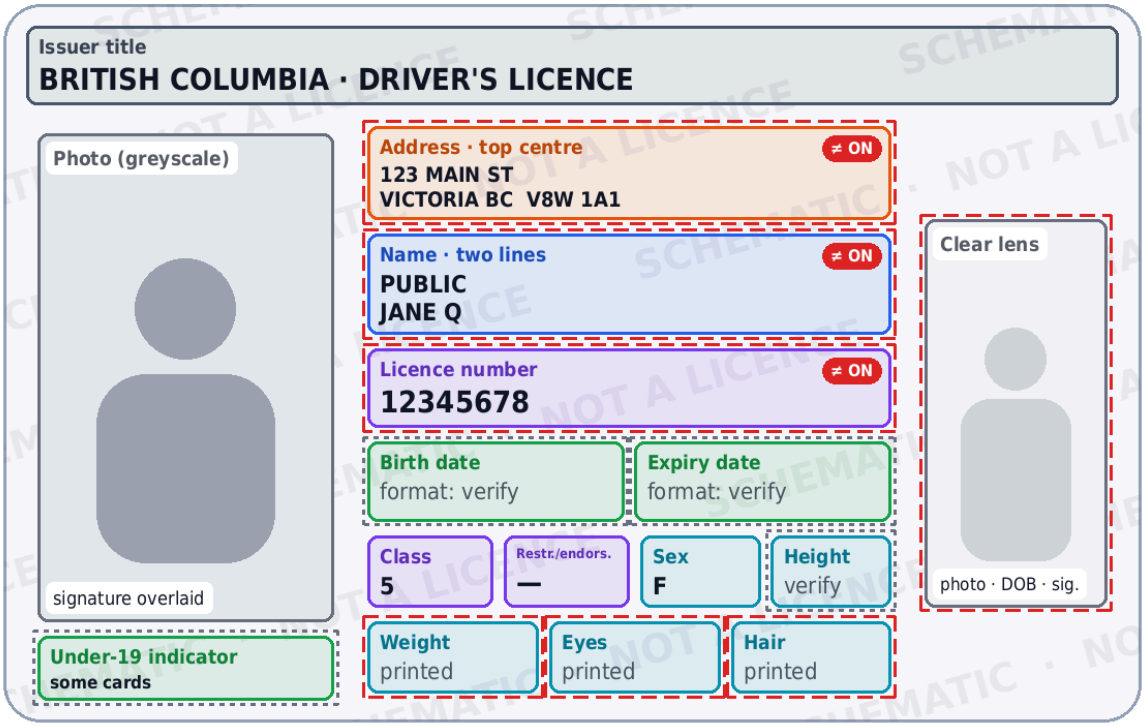


- Field numbers, French labels and the back's two barcodes come from SAAQ's presentation of the 2023 card (SAAQ).
- Number = surname initial + 4 digits + birth date DDMMYY + 2 digits (Sonnet). SAAQ's specimen L1531-171274-08 belongs to a holder born 1974-12-17, so validate both.
- The 2015–2023 card uses the same numbering but lists class 4C (taxi) and a different set of condition codes.

British Columbia

British Columbia · front

Schematic field map · not a licence · 2022+ design, changes per ICBC · positions indicative · fictitious values



What differs from Ontario

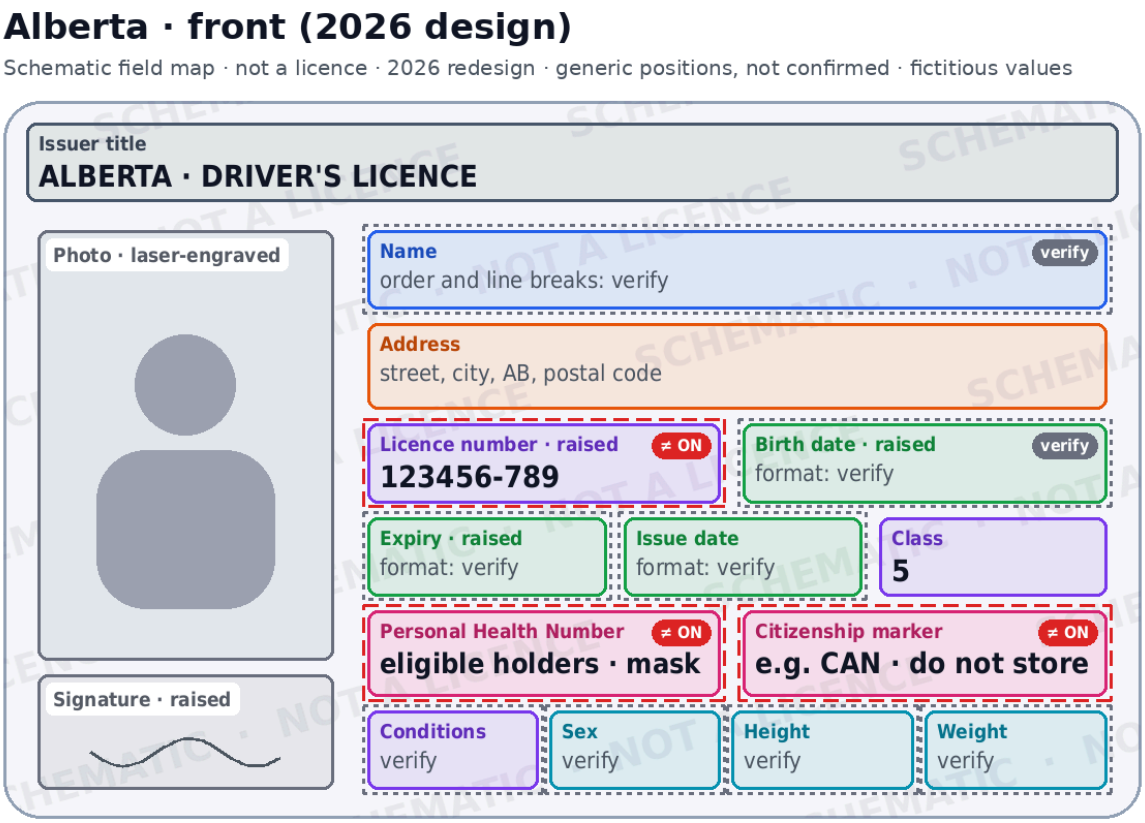
- Number is 7 or 8 digits with nothing embedded, so no surname or birth-date check is possible.
- Surname on line 1, given names on line 2; address sits at the top centre on 2022+ cards.
- Prints weight, eye colour and hair colour. Gloss finish; the clear lens shifts when tilted, so never read data from it.
- The 2017–2022 design is still in circulation with a different field order: train both.

Issuer title (classifier cue) · Name · Dates · Licence data · Address · Physical descriptors · Photo / signature · Differs from Ontario · Not confirmed: verify on specimen



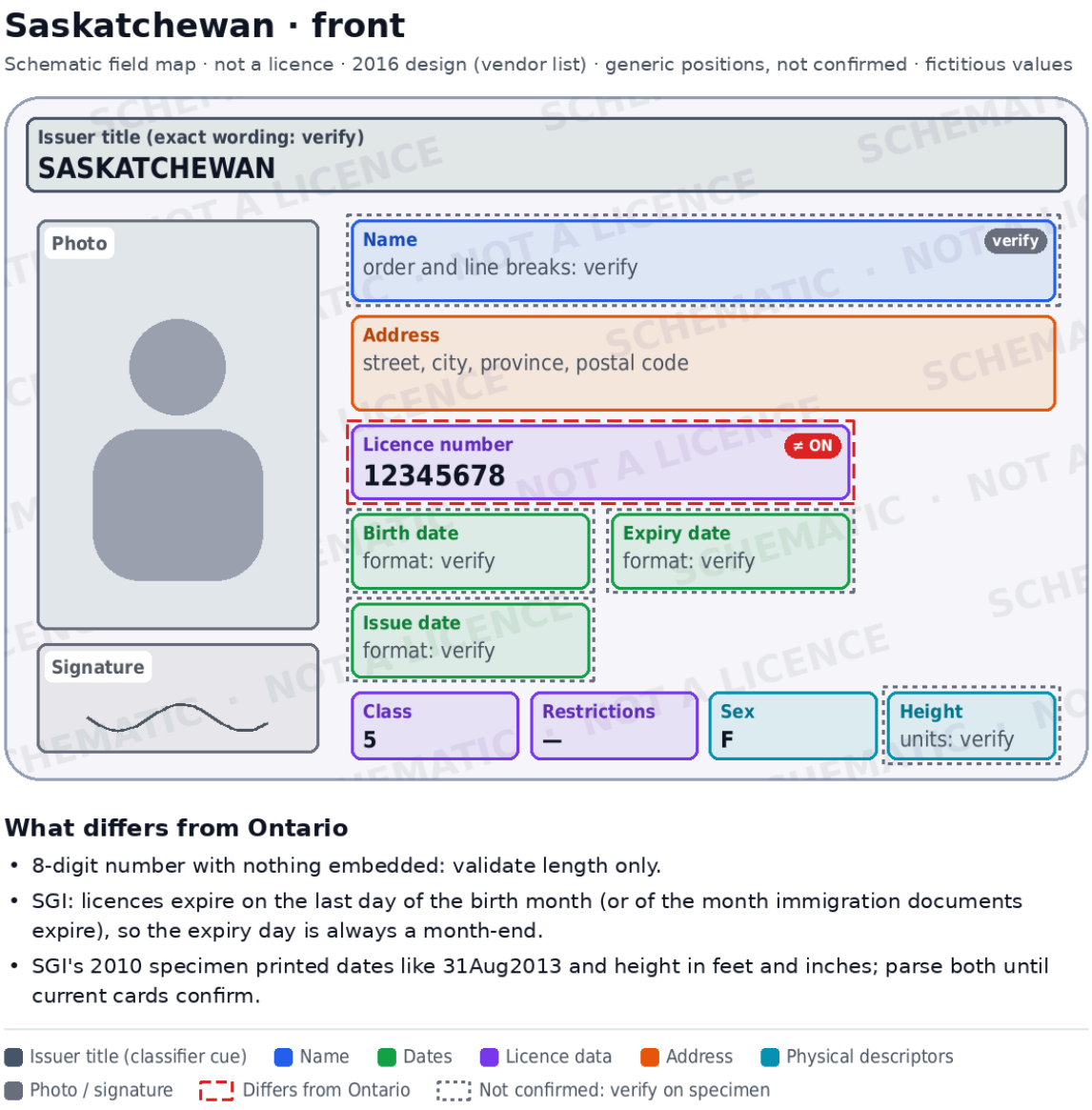
- 7 or 8 digits with nothing embedded (Sonnet).
- Last name on line 1, given names on line 2; sex, eye colour, hair colour, weight and height on the front; a vertical 1D barcode on the back is the card serial (ICBC).
- 2022 cards moved the address to the top centre, reordered the data fields at the bottom centre, and put the secondary photo, full birth date and signature in a clear lens (ICBC card changes).
- Learner, novice and combined licence/Services Card versions are separate templates; the combined card carries the PHN (ICBC).

Alberta



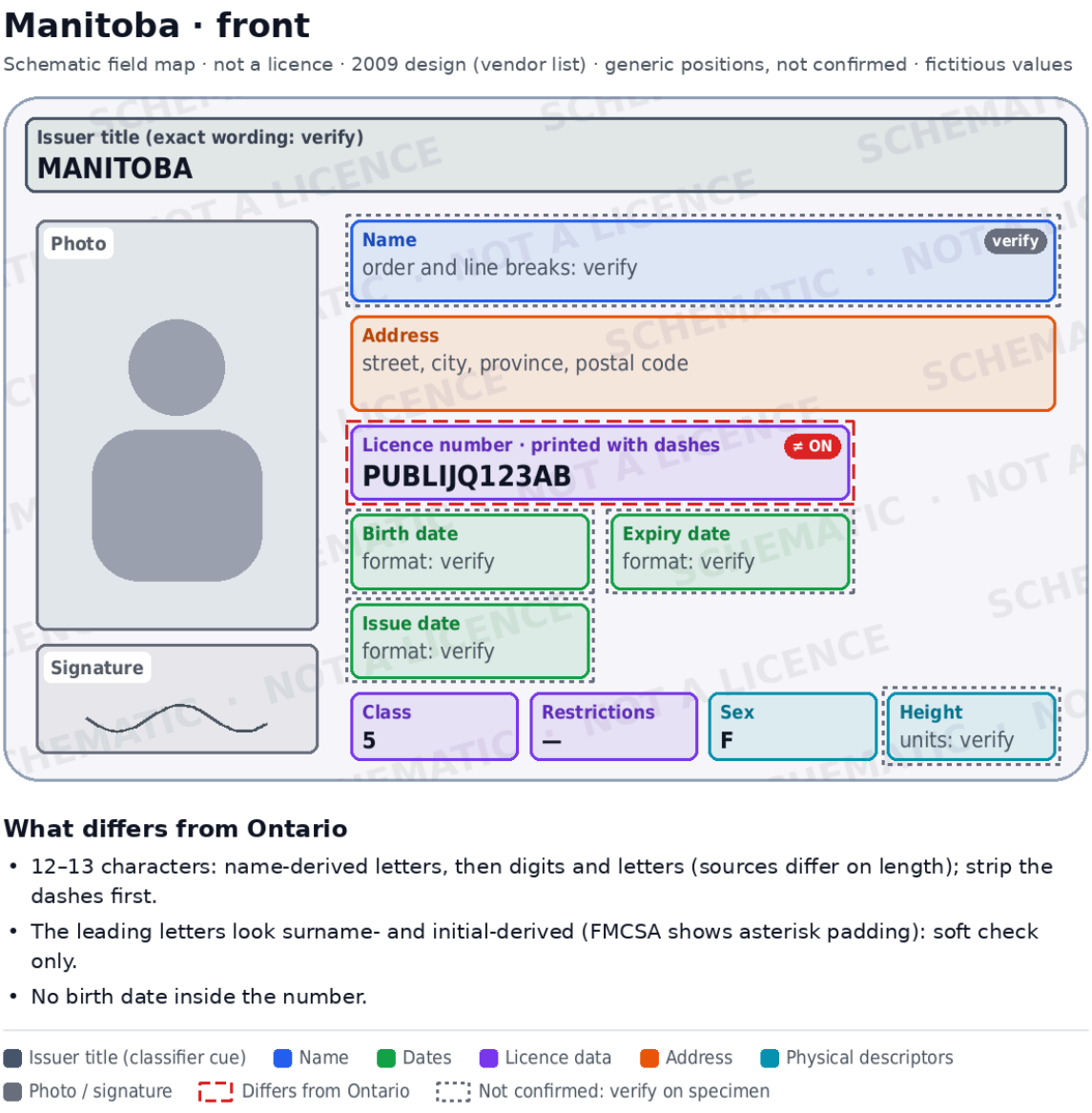
- The 2026 redesign is available now and issued at renewal; current cards stay valid until they expire (Alberta security features). From July 2, 2026 cards can show a PHN and a Canadian citizenship marker (Alberta). Positions aren't published, so the card above is generic.
- Raised number, birth date, expiry date and signature, plus a ghost birth date, look-through windows and laser engraving (Alberta security features).
- Numbers: 123456-789 is most common; A12345, AB1234 and 123456 also exist (Sonnet). Pre-2026 cards print dates as DD MMM YYYY and weight in kg (vendor table).

Saskatchewan



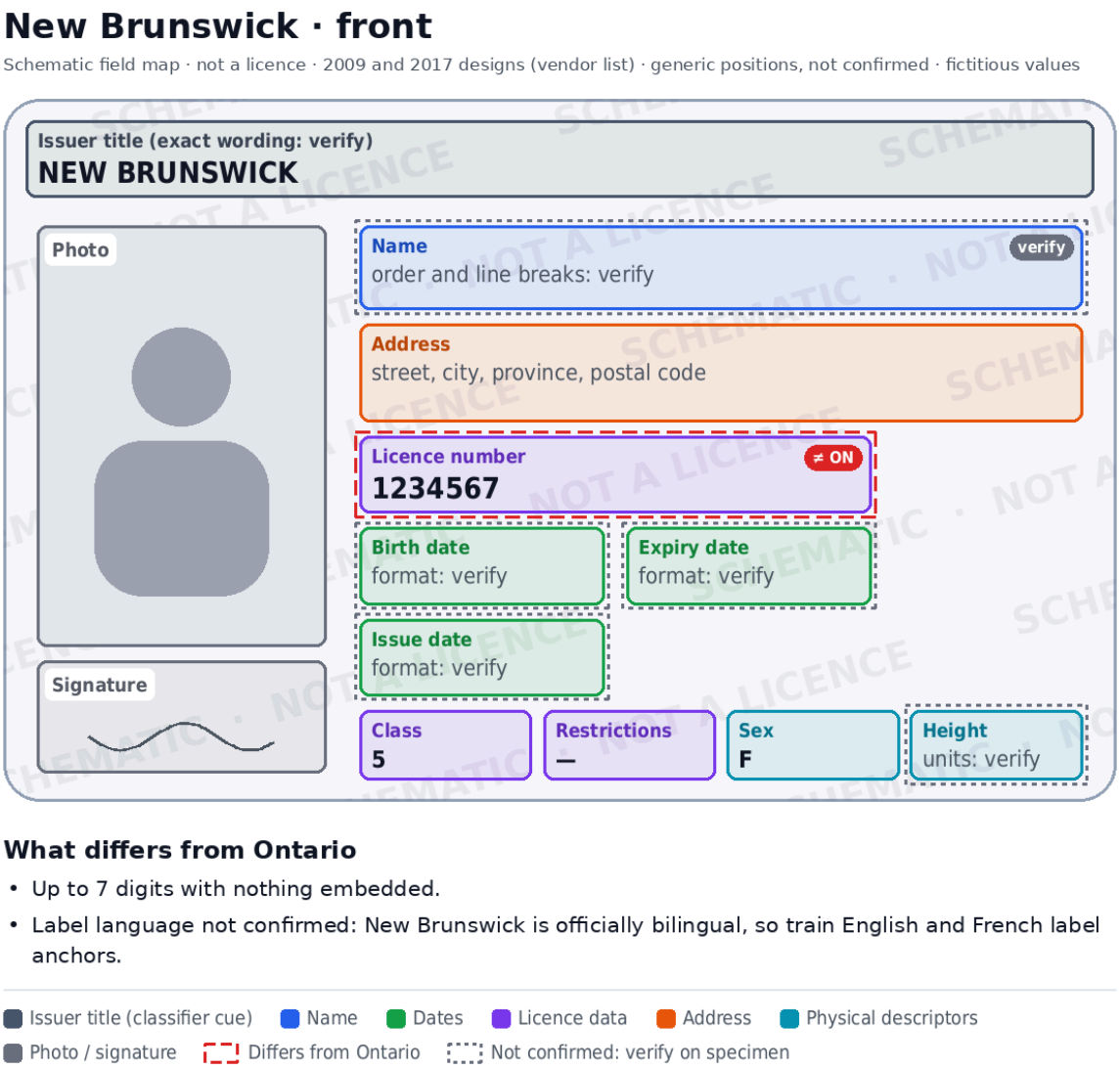
- 8 digits (Sonnet).
- Licences expire on the last day of the birth month, or of the month the holder's immigration documents expire (SGI): the expiry day should always be a month-end.
- SGI's 2010 specimen printed dates like 31Aug2013 and height as 5′ 11″ (SGI 2010); support both until current captures confirm.
- Interim licences are plain white paper (Alberta manual).

Manitoba



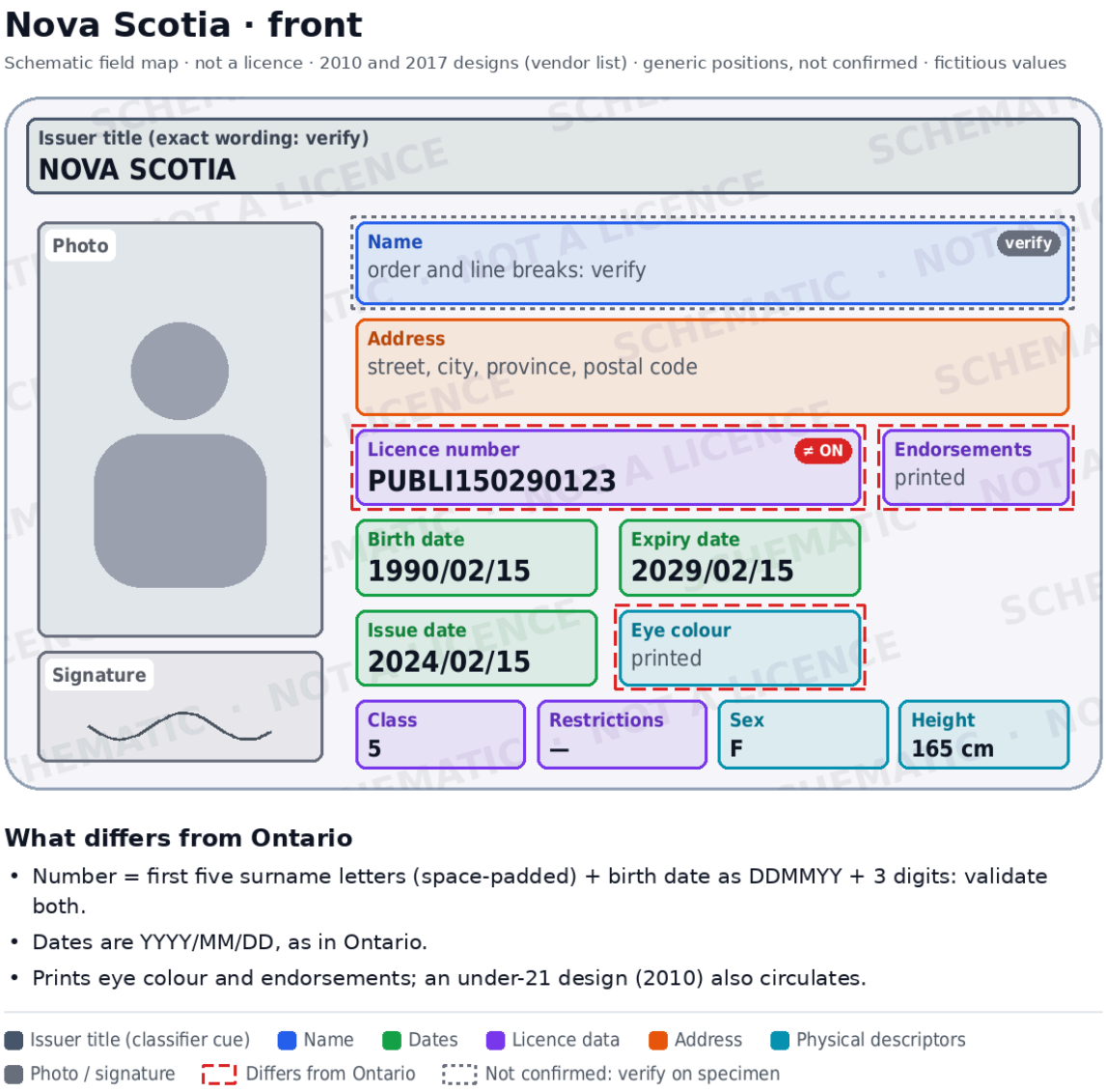
- Sources differ: AAAAAAA###BB entered without dashes (Sonnet) versus 13 characters with asterisk padding (FMCSA). Accept both.
- One design in vendor lists (2009) (Incode).

New Brunswick



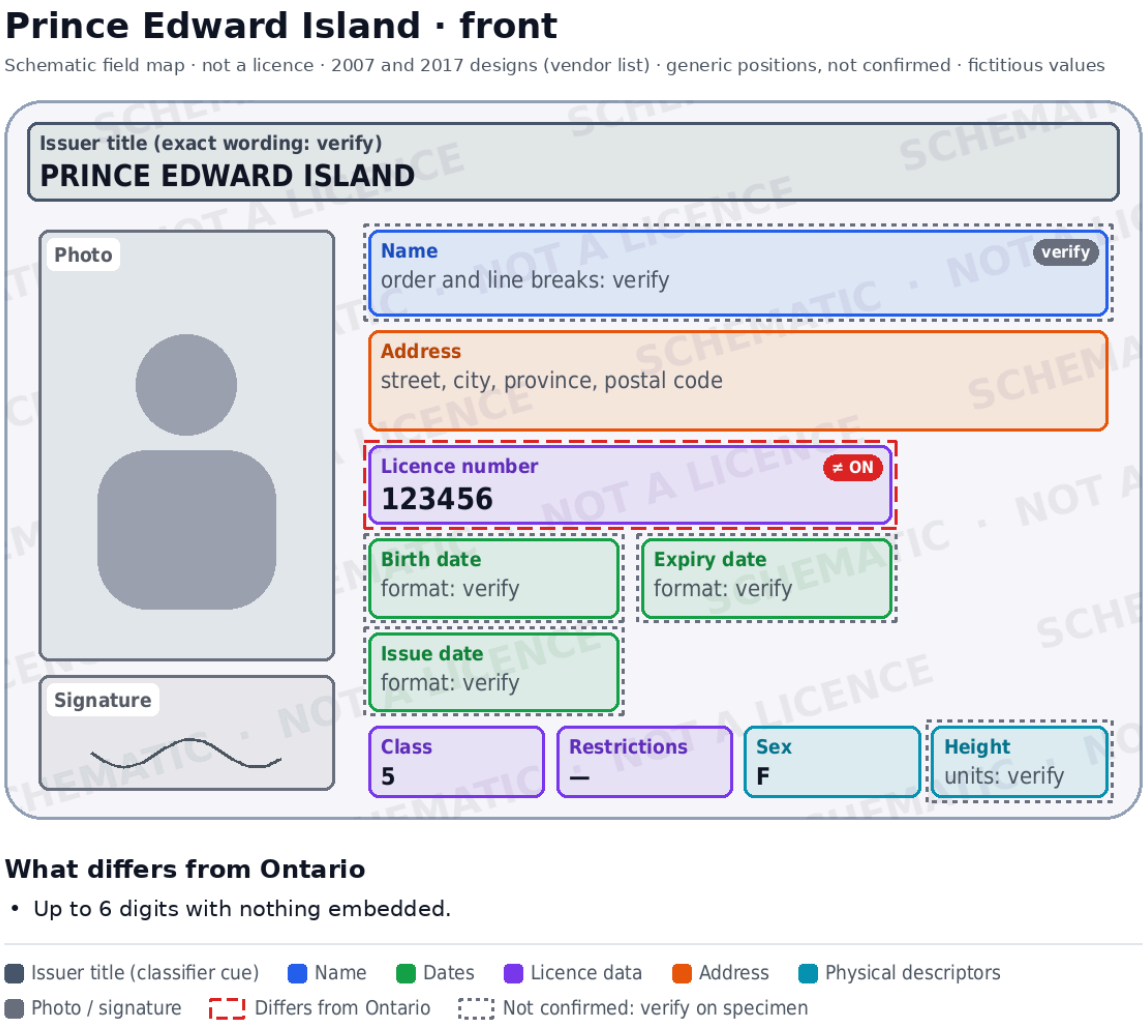
- Up to 7 digits (Sonnet); designs from 2009 and 2017 (Incode).
- Label language not confirmed: train English and French anchors.

Nova Scotia



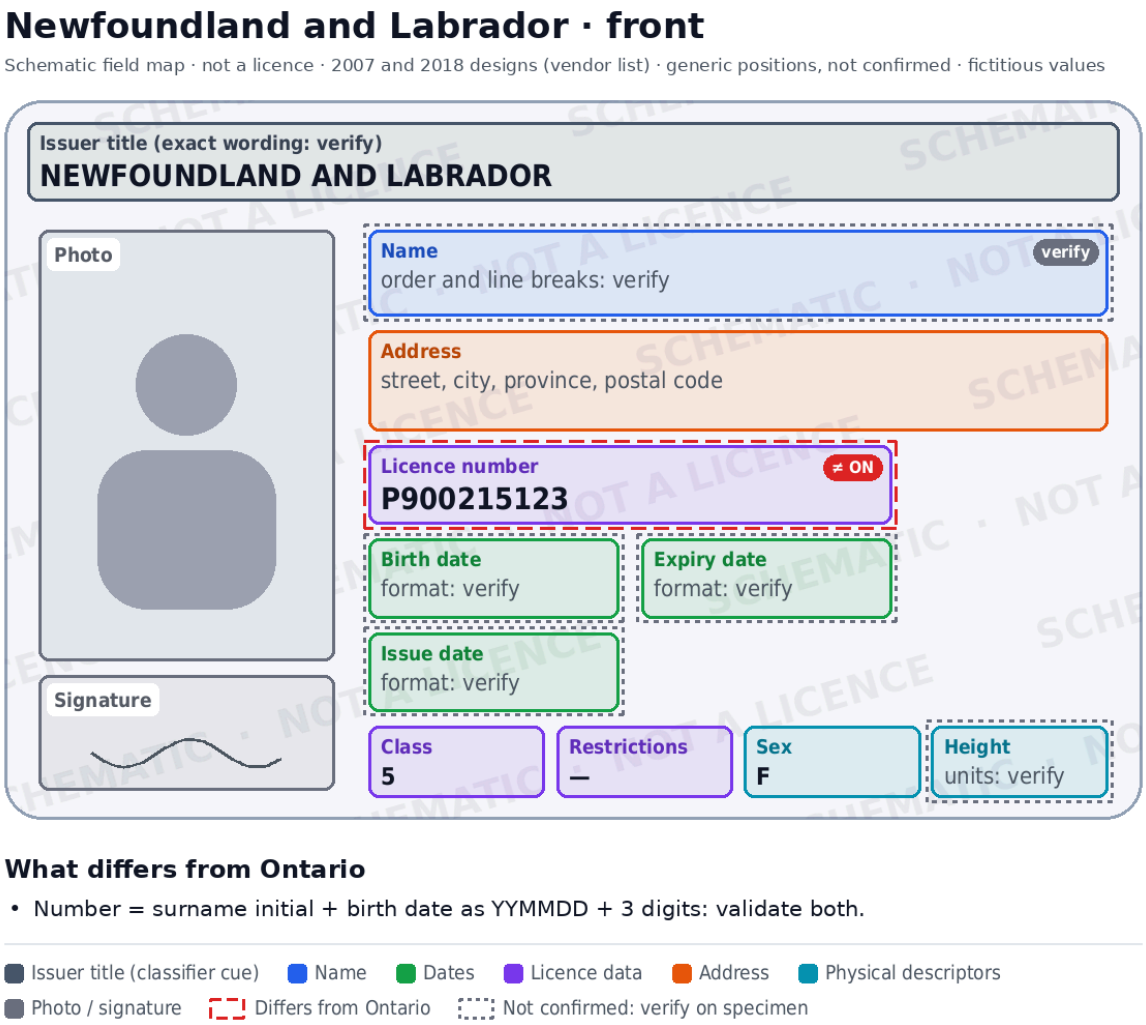
- First five surname letters, space-padded, + DDMMYY + 3 digits (Sonnet); FMCSA shows asterisk padding instead, so accept both.
- Dates YYYY/MM/DD; eye colour and endorsements printed; document reference of two letters + 20 digits (vendor table).
- An under-21 design (2010) also circulates (Incode).

Prince Edward Island



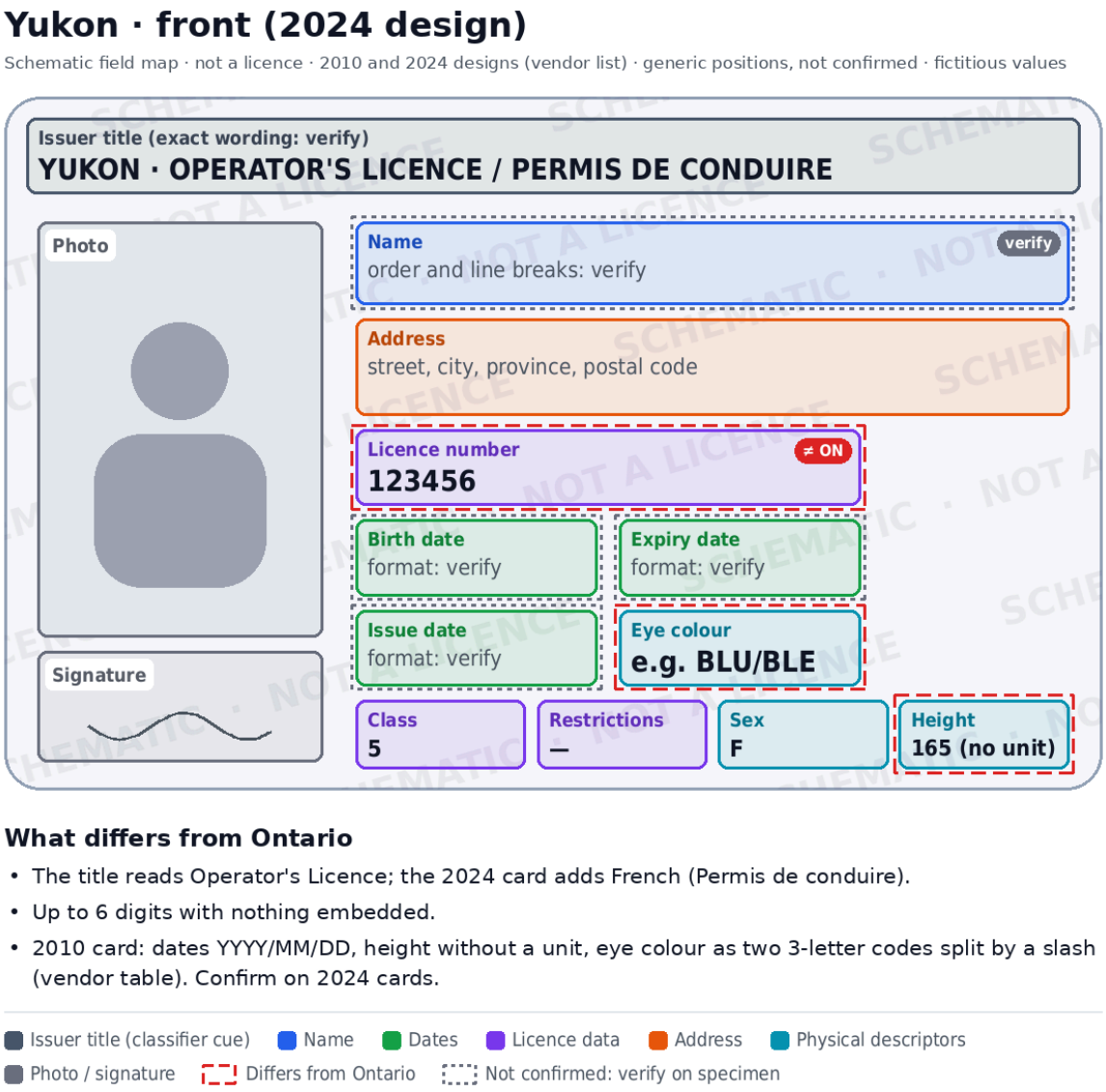
- Up to 6 digits (Sonnet); designs from 2007 and 2017 (Incode).

Newfoundland and Labrador



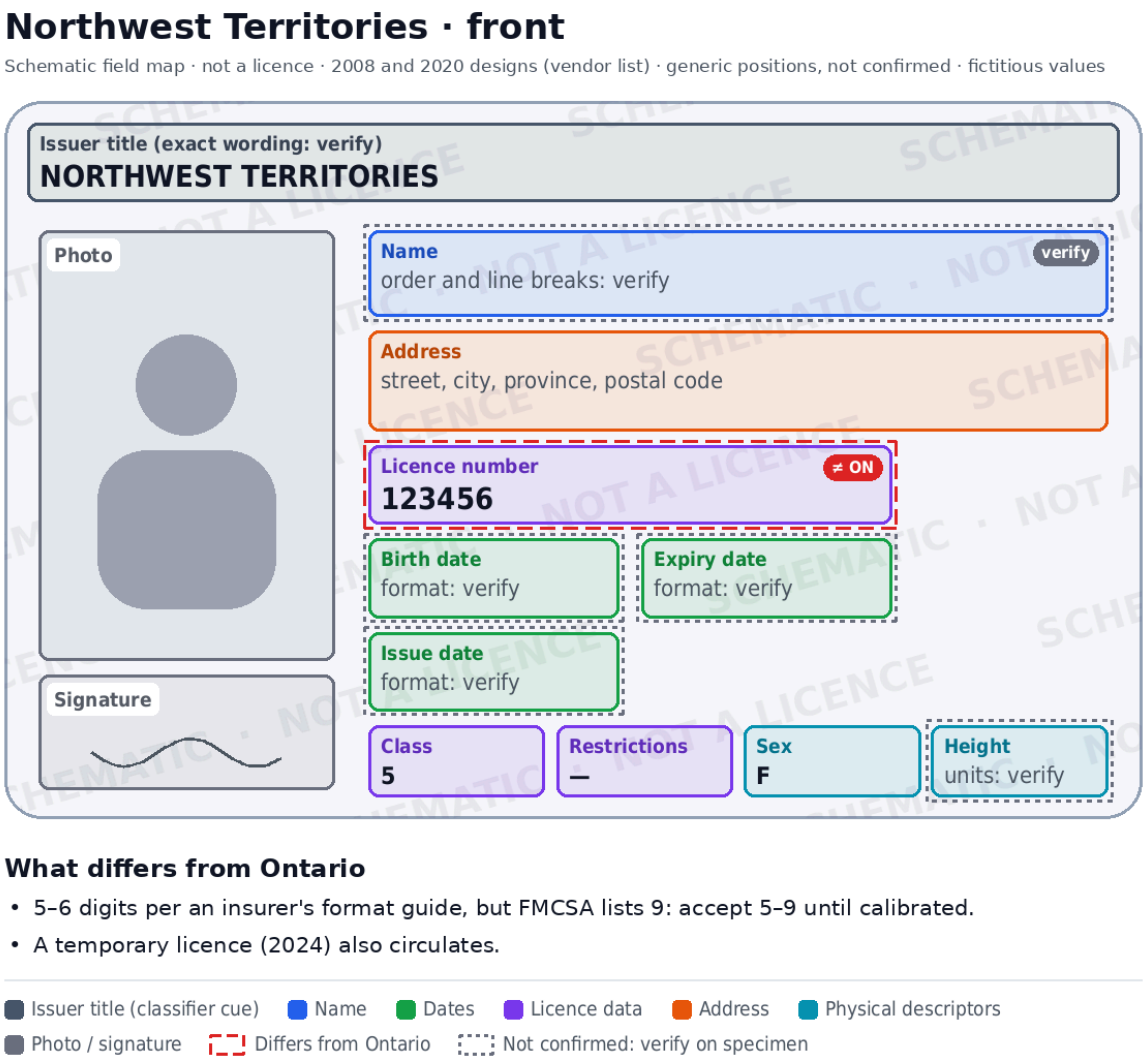
- Surname initial + YYMMDD + 3 digits (Sonnet): validate both; designs from 2007 and 2018 (Incode).

Yukon



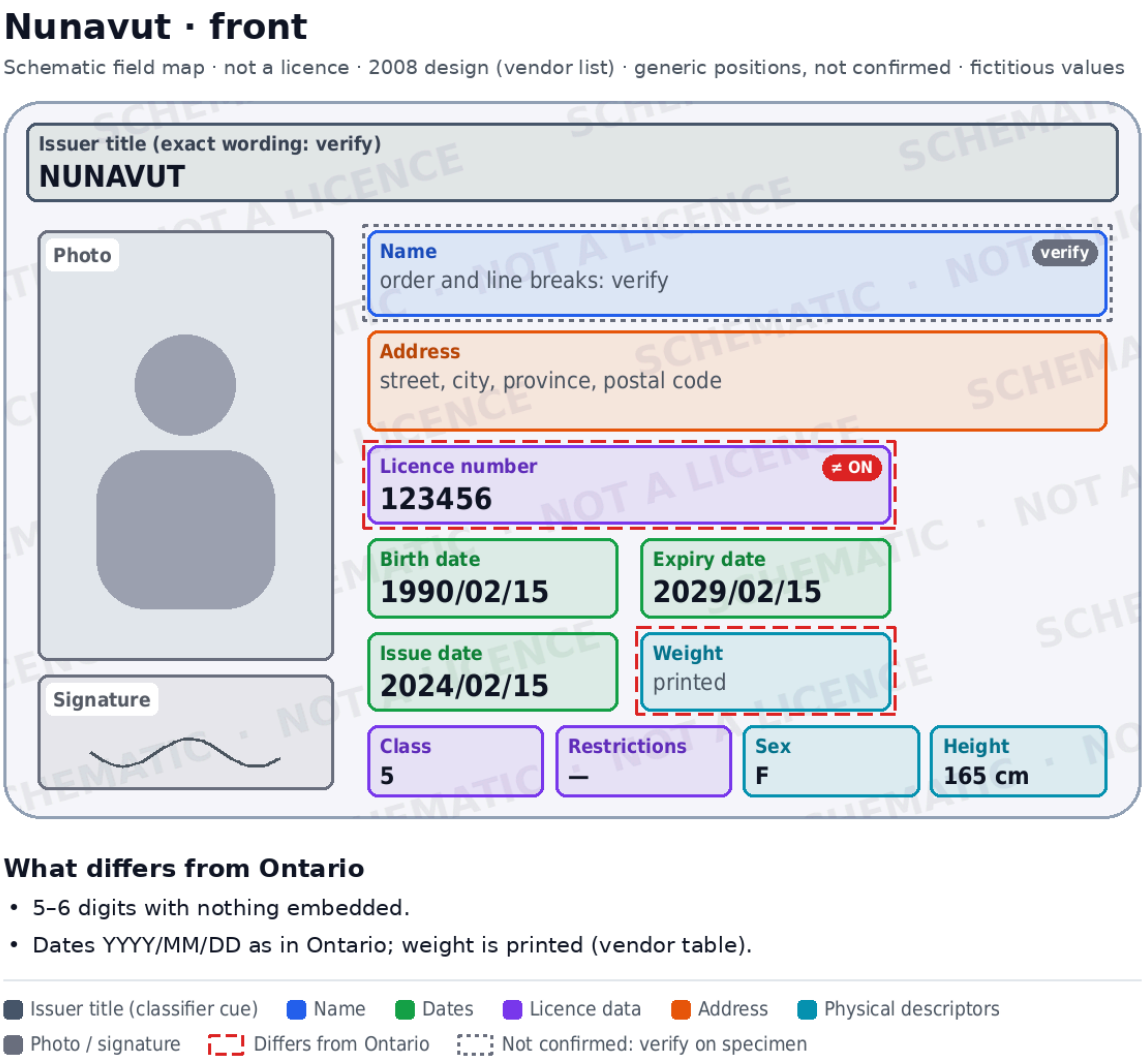
- Up to 6 digits (Sonnet). The 2024 card is titled Permis de conduire (Operator's Licence) (Incode).
- On the 2010 card: dates YYYY/MM/DD, height without a unit, eye colour as two 3-letter codes split by a slash (vendor table).

Northwest Territories



- 5 or 6 digits (Sonnet), but FMCSA lists 9: accept 5–9 until calibrated. Designs from 2008 and 2020, plus a 2024 temporary licence (Incode).

Nunavut



- 5 or 6 digits (Sonnet); dates YYYY/MM/DD and weight printed (vendor table); one design (2008) (Incode).

Master comparison matrix

Every jurisdiction differs from Ontario in its number pattern, and only four embed the date of birth in the number: Ontario, Quebec, Nova Scotia, and Newfoundland and Labrador. "Not confirmed" means no public source states it; take dates from the barcode until your own calibration captures settle the printed format.

Jurisdiction	Number as printed	Regex (separators removed)	Birth date inside number	Printed date format	Label language	Card versions in circulation
Ontario (ON)	A1234-56789-01234	`^[A-Z]\d{14}$`	YYMMDD in digits 9–14	YYYY/MM/DD	English/French	2011; enhanced 2010
Quebec (QC)	A1234-123456-12	`^[A-Z]\d{12}$`	DDMMYY in digits 5–10	YYYY-MM-DD	French only	2015; 2023
British Columbia (BC)	1234567 or 12345678	`^\d{7,8}$`	No	Not confirmed	English	2020, 2022, 2025; enhanced 2009
Alberta (AB)	123456-789; older A12345, AB1234, 123456	`^\d{6,9}$` or `^[A-Z]{1,2}\d{4,5}$`	No	DD MMM YYYY on pre-2026 cards; 2026 card not confirmed	English	2009, 2018, 2026
Saskatchewan (SK)	12345678	`^\d{8}$`	No	Not confirmed	English	2016
Manitoba (MB)	Name-derived letters, then digits; printed with dashes	`^[A-Z][A-Z*]{4}[A-Z][A-Z*][A-Z0-9]{5,6}$`	No	Not confirmed	Not confirmed	2009
New Brunswick (NB)	Up to 7 digits	`^\d{1,7}$`	No	Not confirmed	Not confirmed	2009, 2017
Nova Scotia (NS)	AAAAADDMMYY### (surname letters, space-padded)	`^[A-Z][A-Z* ]{4}\d{9}$`	DDMMYY in characters 6–11	YYYY/MM/DD	English	2010, 2017; under-21 2010
Prince Edward Island (PE)	Up to 6 digits	`^\d{1,6}$`	No	Not confirmed	English	2007, 2017
Newfoundland and Labrador (NL)	AYYMMDD###	`^[A-Z]\d{9}$`	YYMMDD in characters 2–7	Not confirmed	English	2007, 2018
Yukon (YT)	Up to 6 digits	`^\d{1,6}$`	No	YYYY/MM/DD (2010 card)	English; 2024 card adds French	2010, 2024
Northwest Territories (NT)	5–6 digits (FMCSA lists 9)	`^\d{5,9}$`	No	Not confirmed	Not confirmed	2008, 2020; temporary 2024
Nunavut (NU)	5–6 digits	`^\d{5,6}$`	No	YYYY/MM/DD	Not confirmed	2008

Sources: number formats from Sonnet and FMCSA; dates from Tungsten field tables and SAAQ; versions from Incode, SAAQ and Alberta.

Back-of-card barcode (AAMVA PDF417)

Decode the back before you OCR the front. Every Canadian issuer has an AAMVA issuer identification number (IIN) that sits in the barcode header, so one decode identifies the jurisdiction and returns clean field values to check the front against (AAMVA).

Jurisdiction	IIN
Ontario	636012
Quebec	604428
British Columbia	636028
Alberta	604432
Saskatchewan	636044
Manitoba	636048
New Brunswick	636017
Nova Scotia	636013
Prince Edward Island	604426
Newfoundland and Labrador	636016
Yukon	604429
Northwest Territories	604434
Nunavut	604433

Data elements to map into the normalized record (AAMVA DL/ID Card Design Standard):

Element	Field	Note
DAQ	Licence number	Compare with the front, separators removed
DCS / DAC / DAD	Family, first, middle names	
DBB / DBD / DBA	Birth, issue, expiry dates	Canadian issuers use CCYYMMDD; U.S. issuers MMDDCCYY
DBC	Sex	1 = male, 2 = female, 9 = not specified

Element	Field	Note
DAU / DAY	Height, eye colour	
DAG / DAI / DAJ / DAK	Street, city, province, postal code	
DCF	Document discriminator	Ontario's DD/REF
DCA / DCB / DCD	Class, conditions, endorsements	Jurisdiction-specific codes
DCG	Country	CAN

- Detect date order from the value rather than trusting the header: if the first four digits are a plausible year, it is CCYYMMDD.
- Old barcodes can carry a five-digit IIN without the leading 6 (LEADTOOLS); normalize before lookup.
- Quebec's back has two barcodes: a 1D code holding only the licence number and a 2D code holding everything except photo and signature (SAAQ).
- BC's enhanced licence has an ID barcode under the magnetic stripe and no 2D barcode; fall back to front OCR (ICBC).
- Decoders: ZXing, or the on-device barcode APIs in Google ML Kit and Apple Vision; ML Kit also returns parsed AAMVA licence fields.

Training plan for the OCR model

Don't train one end-to-end model per province. Train three small models and let per-jurisdiction rules (regex, date parsing, number checks) do the rest; adding a province or a redesign then means new templates and labels, not new architecture.

1. **Card detector:** find the card's four corners and warp it to the ID-1 shape (85.6 × 54 mm, about 1.586:1).
2. **Classifier:** one class per jurisdiction × card version × side, for example `ON-2011-front`, `QC-2023-back`, `AB-2026-front`. Auto-label backs from the barcode IIN.
3. **Field reader:** start with zone templates plus a text recognizer for Ontario (fastest to ship), then move to a layout-aware key–value model so redesigns such as Alberta 2026 need relabelling, not new zones. Check model licences before shipping; some popular document models' weights are non-commercial.

Label taxonomy (one box per item): `family_name`, `given_names`, `name_full`, `address_line`, `city_prov_postal`, `licence_number`, `dob`, `issue_date`, `expiry_date`, `sex`, `height`, `weight`, `eye_colour`, `class`, `conditions`, `endorsements`, `document_reference`, `age_banner`, `label_anchor`, `photo`, `ghost_photo`, `signature`, `barcode_2d`, `barcode_1d`, plus `health_number` and `citizenship_marker` so the model can find and mask them.

Data sources:

- Official specimens to build templates and zone maps (linked in Sources).
- Consented captures from staff and beta users, with written consent and a deletion date.
- Synthetic text crops: fictitious values rendered in similar fonts on plain backgrounds to train the recognizer. Keep synthetic data at field-crop level; don't generate realistic full-card images.

Augmentations that match real phone captures: glare and specular highlights (glossy and laser-engraved cards), motion and defocus blur, low light and sensor noise, rotation and perspective, JPEG compression, fingers over edges, holographic overlays.

Evaluate with:

- Exact-match rate per field per jurisdiction after normalization, and character error rate for names and addresses.
- Barcode decode rate, classifier confusion matrix, and the share of submissions sent to manual review.
- A test split by person, not by image, sliced by jurisdiction, card version, device and lighting.

Before committing to a build, compare the labelling effort with commercial SDKs; vendors such as Tungsten and Incode already list all 13 jurisdictions.

Edge cases to handle from day one

These variants reach production in the first weeks; give each a class in the classifier and a rule in the review queue.

- **Older designs stay valid until expiry.** BC cards issued 2017–January 2022 remain in circulation (ICBC), and Alberta's pre-2026 cards stay valid until they expire (Alberta).

- **Interim paper licences.** Ontario issues temporary licences before the card arrives (DriveTest); Saskatchewan interims are plain white paper (Alberta exchange manual); Alberta issues an interim receipt for a 60-day grace period (Alberta). Decide whether to reject these or route them to manual review.
- **Enhanced licences.** BC's program is discontinued but cards circulate until expiry; they have no 2D barcode and carry a 3-line passport-style MRZ on the back instead (ICBC). Ontario also has a 2010 enhanced card (Incode). Add an MRZ parser as a fallback.
- **Health numbers and citizenship markers.** BC's combined licence and Services Card prints the Personal Health Number on the back and puts it in the magstripe and the 2D barcode (ICBC). Alberta cards issued from July 2, 2026 can show a PHN and a "CAN" citizenship marker (Alberta). Mask them in images and drop them from parsed data.
- **Learner and novice cards.** BC issues separate learner and novice cards (ICBC); Ontario's G1, G2, M1 and M2 appear in the class field. Decide which classes your app accepts.
- **Age indicators.** Ontario cards issued at 16–18 carry an "AGE 19" banner and date (AGCO); BC prints an under-19 indicator under the signature (ICBC); Nova Scotia has an under-21 design (Incode).
- **Language and accents.** Quebec labels are French only (SAAQ); Ontario's are bilingual; Yukon's 2024 card is titled in French and English (Incode). Ontario prints French accented characters such as ç, è, é, ê and ë in names (Ontario), so train the recognizer on accented capitals and keep them.
- **Raised and laser-engraved text.** Ontario raises the licence number and date of birth (AGCO); Alberta's 2026 card raises the number, birth date, expiry date and signature (Alberta); BC's 2022 card is glossy (ICBC). Expect shadows and glare on exactly the fields you most need.
- **Quebec validity.** The Quebec card states that payment is required every year on the holder's birthday (SAAQ), so a future expiry date alone doesn't prove the licence is in good standing.

Privacy and compliance

Canada's privacy commissioners treat examining, recording and scanning a licence as collecting personal information, and expect you to state the purpose, take the least information that meets it, and protect what you keep; consent does not remove that duty

(OPC, Alberta and BC commissioners). Four laws apply depending on where you operate: PIPEDA federally, Alberta's and BC's PIPAs, and Quebec's private-sector law. This is engineering guidance, not legal advice; have counsel confirm your design.

- **Treat the number as sensitive.** Licence numbers rarely change and are valuable to identity thieves, and some embed year of birth, gender or part of the surname (OPC). Don't use it as your user ID; for duplicate detection store a keyed hash (HMAC) instead.
- **Keep the result, not the card.** The commissioners say photocopying or scanning generally goes too far for most purposes. Store the verification outcome, the fields you need (name, date of birth, expiry, licence class) and a reference to the check; delete card images once review is done.
- **Drop what you never needed.** Mask health numbers and citizenship markers in images and discard them from parsed data (BC combined cards, Alberta 2026 cards); skip physical descriptors unless you use them.
- **Biometrics need explicit consent.** If you compare the selfie with the card photo, ask for consent at that step, state retention, and delete face templates after the match. Quebec has specific rules for biometric verification; check them before launching there.
- **Security basics.** Encrypt images and fields at rest, restrict reviewer access, log every view, and publish a retention schedule.

Sources

Calibrate against the official specimens first; vendor tables fill gaps but can lag a redesign.

Issuer specimens and guidance:

- AGCO: Tips for checking ID (Ontario specimen, front and back)
- Ontario: Change information on a driver's licence
- Ontario Design System: licence number hint text
- DriveTest: Ontario licences
- SAAQ: Presentation of Québec driver's licences
- ICBC: Card security features (MV2804A)
- ICBC: Card feature changes, 2022
- Alberta: New driver's licence and ID cards

- Alberta: Driver's licence security features
- Alberta: Saskatchewan licence exchange policy
- SGI: Driver's licence (handbook)
- SGI: One-part driver's licence backgrounder (2010)

Standards and regulators:

- AAMVA: Issuer identification numbers
- FMCSA: Canadian licence number formats
- Privacy commissioners of Canada, Alberta and BC: Collection of driver's licence numbers

Industry references (not official):

- Sonnet Insurance: Driver's licence number formats
- Incode: Supported IDs, North America
- Tungsten field tables: Ontario, Nova Scotia, Alberta, Yukon, Nunavut
- LEADTOOLS: AAMVA ID structure